AI-POWERED CRIMINAL NETWORK ANALYSIS SYSTEM

BUILD WITH BHARAT 2.0 HACKATHON | ULTIMATE ALL-IN-ONE MASTER DOCUMENT

Problem Statement ID: SIH PS 26189

> 📌 **Document Purpose (Sampurna Project Guide)**
> Is single document me project ki poori kahani, simple logic, visual diagrams, data pipelines, real vs dummy sources, UI layout, legal compliance (Section 65B), aur judge Q&A defense included hai. Isko padhne ke baad kisi team member ya judge ko kuch aur padhne ki zaroorat nahi padegi.

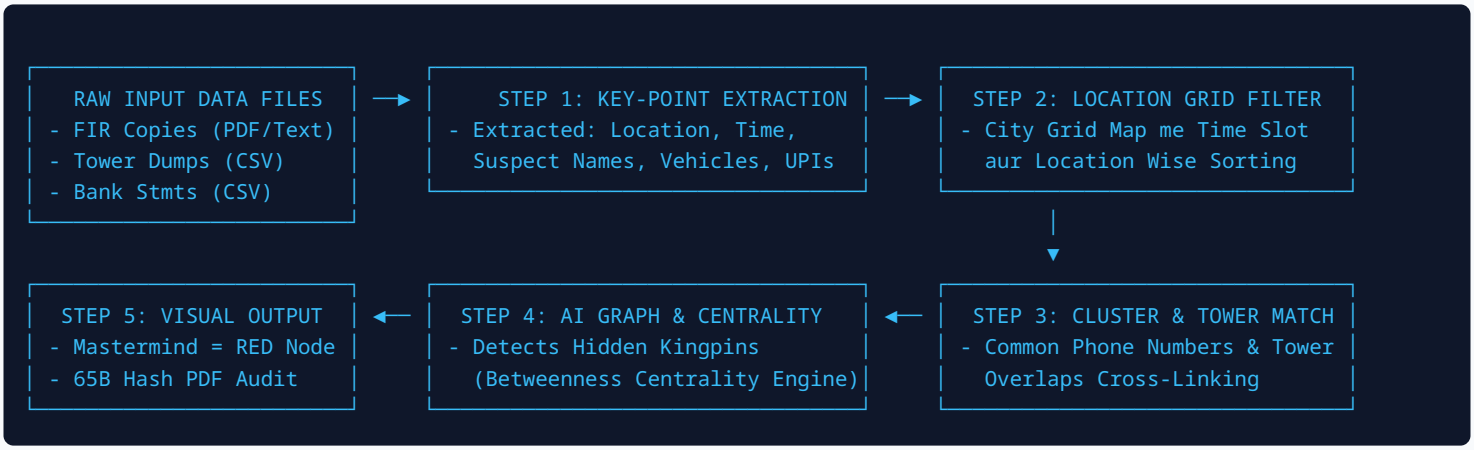
1. Simple Language Me Idea (Police Ki Problem & Hamara Solution)

Asal Problem: Ek sehar me 50 police stations hote hain [cite: 1, 2, 5]. Har station apni FIRs, Call Records (CDR), aur Bank Statements alag-alag files me rakhta hai [cite: 1, 2, 5]. Agar Station-A ki FIR me likha hai ki robbery me *"Black Pulsar"* aur nickname *"Kala"* tha, aur Station-B ki FIR me bhi *"Black Pulsar"* thi—to dono stations ko aapas me pata hi nahi chalta [cite: 1, 2]. In hazaron disconnected files se dots connect karne me police ko 3 se 5 din lag jaate hain [cite: 1, 2, 5].

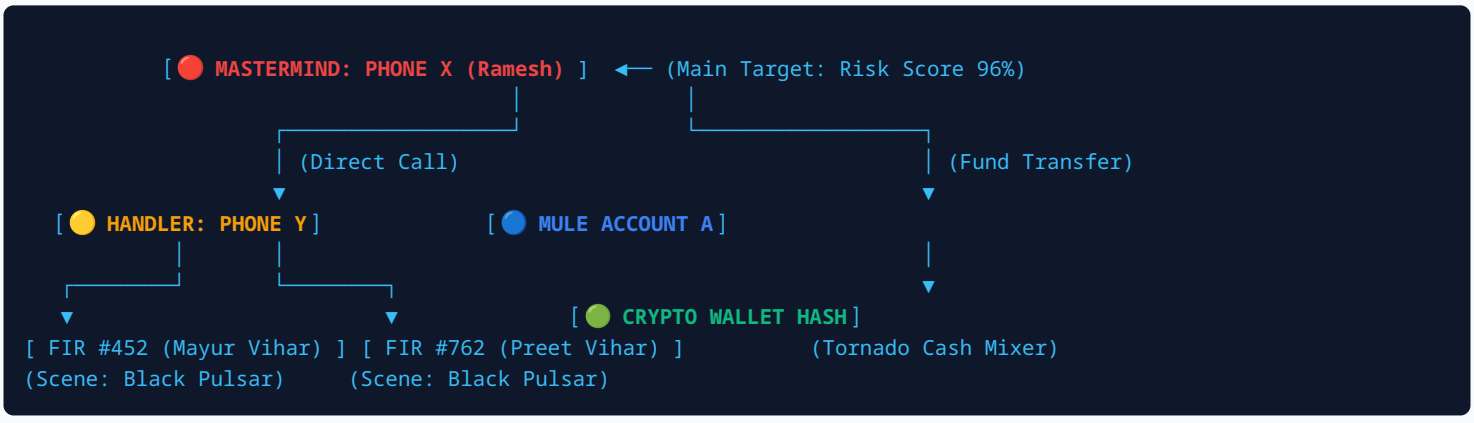
Hamara Solution: Yeh system ek **Centralized AI Brain** ki tarah kaam karta hai [cite: 1, 2, 5]. Yeh 50 stations ki saari Hindi/English FIRs, Phone Call Logs, Bank Transfers, aur Crypto Wallets ko ek saath read karta hai aur screen par ek **Visual Map (Network Graph)** bana deta hai [cite: 1, 2, 5]. Jo criminal sabse peeche chup kar sab control kar raha hota hai (Mastermind), system use **RED PULSE ALERT** ke sath screen par highlight kar deta hai [cite: 1, 2, 5].

2. Visual Flow Diagrams (Data Se Mastermind Tak)

A. System Diagram (Step-by-Step Mechanism)



B. Visual Graph Network Map (Police Screen Par Kaisa Dikh-ega)



3. Complete Features Breakdown (Kya Feature Hai & Kaise Kaam Karega)

Feature Name		Kaise Kaam Karega (Working Principle)

	Simple Words Me Matalab	
1. OCR & Multilingual FIR Parser	Scanned FIR photo/PDF se text padhna.	Hindi aur Hinglish FIR copies se OCR (Tesseract) text extract karta hai, aur NLP engine usme se Location, Vehicle No, Suspect Alias alag nikal leta hai [cite: 1, 2, 5].
2. Cross-Station Cluster Engine	Alag-alag station ki FIRs ko jodna.	Agar Mayur Vihar FIR aur Preet Vihar FIR me "Black Pulsar" + "Raat 11 PM" common hai, to dono ko ek single "Snatching Racket Cluster" me daal deta hai [cite: 1, 2].
3. Tower Dump & CDR Cross-Matcher	Crime spot par active phones dhoondhna.	10,000 call logs me se search karta hai ki kaun sa phone number exact crime time par multiple crime locations par active tha [cite: 2].
4. AI Mastermind Detector	Asli Leader ko parda-fash karna.	Betweenness Centrality Algorithm dekhta hai ki kis banday ne crime spot par na hote hue bhi shooters ko call ki aur money distribute ki [cite: 1, 2, 5]. Use RED Node bana deta hai [cite: 2].
5. Hybrid Crypto-Fiat Tracker	Bank + Crypto transfer track karna.	Bank Accounts se lekar Crypto Wallet Hashes tak paisse ke raste ko map karta hai [cite: 5].
6. Section 65B Legal Hash Logger	Court ke liye saboot pakka karna.	Har uploaded file ka SHA-256 Hash generate karke immutable log banata hai, jisse evidence tampered na ho [cite: 5].
7. Human-in-the-Loop Override	Police ko full control dena.	Police officer galat link Reject kar sakta hai, suspect ki photo edit kar sakta hai, aur custom PDF report export kar sakta hai [cite: 2, 5].

4. Real World Vs Hackathon Data Strategy

Data Type	Real Police System (Production)	Hackathon Demo Strategy (Dummy Data)
Phone CDR & Tower Logs	Telecom Providers (Airtel, Jio) via Police Warrants & CEIR Portal [cite: 1].	Python `Faker` script generating mock Delhi-NCR Cell Towers & CDR CSVs [cite: 1, 5].
Bank Statements & Fraud	Financial Intelligence Unit (FIU-IND) & 1930 Portal APIs [cite: 1].	Anonymized Kaggle `PaySim` Mobile Money Fraud Dataset [cite: 1, 5].
FIR Text Copies	CCTNS (Crime & Criminal Tracking Network) Government Database [cite: 1].	Synthetic Hindi/English FIR PDFs generated via OpenStreetMap coordinates [cite: 1, 5].

5. Market Comparison & USP (Pehle Se Kaise Alag Hai?)

- **Palantir Gotham (US):** High-end military tool [cite: 1, 5]. Very expensive hai aur Indian Hindi/Hinglish FIRs handle nahi kar sakta [cite: 1, 5]. Our system is open-source & Indian context ready [cite: 1, 5].
- **IBM i2 Analyst's Notebook:** Offline software [cite: 1, 5]. Isme har entry manually karni padti hai [cite: 1, 5]. Our system automatically reads FIRs and auto-syncs across stations [cite: 1, 5].
- **CCTNS (India):** Sirf static file cabinet ki tarah data store karta hai [cite: 1, 2, 5]. Our system AI Analyst ki tarah dots connect karta hai [cite: 1, 2, 5].

6. Highly Effective Team & Judge Q&A Defense Strategy

Q1: Hackathon me real police data nahi milta, to aapka system demo me kya dikhayega?

Ans: Humne real-world schema ko replicate karne wala Python Synthetic Generator banaya hai jo Delhi-NCR coordinates par mock CDRs aur FIRs generate karta hai [cite: 1, 5]. Judges System Architecture, NLP Entity Extraction, aur Neo4j Graph Logic test karte hain, private real data nahi [cite: 1, 5].

Q2: Criminal fake naam bata de (FIR-1 me 'Ramesh', FIR-2 me 'Kala'), to system use ek kaise samjhega?

Ans: System **Multi-Factor Matching Score** nikalta hai: `(Name Match 50%) + (Phone/IMEI Overlap 100%) + (Tower Location Overlap 90%)` [cite: 1, 2, 5]. Naam alag hone par bhi agar phone number ya IMEI same hai, to system auto-link kar deta hai [cite: 1, 2, 5].

Q3: Cell tower dump me 10,000 aam janta (innocent citizens) hote hain, kya AI sabko criminal dikha dega?

Ans: Bilkul nahi! System me **Frequency & Noise Filter** hota hai [cite: 2, 5]. Jo banda 1 baar tower par tha, wo auto-drop ho jata hai [cite: 2]. Sirf wahi numbers highlight hote hain jo multiple crime spots par common the aur jinki call history suspects se linked hai [cite: 2].

Q4: Scanned ya gandi handwriting wali PDF FIRs ko AI kaise padhega?

Ans: Hum **OCR (Optical Character Recognition / Tesseract Engine)** use karte hain jo image ko digital text me badalta hai, aur Multilingual IndicBERT model usme se key details JSON format me extract karta hai [cite: 1, 2, 5].

Q5: Court me yeh evidence kaise valid hoga? Digital evidence me tampering ka risk rehta hai.

Ans: System har uploaded file ka instant **SHA-256 Cryptographic Hash** create karke log kar deta hai [cite: 5]. Court report me timestamped certificate milta hai jo **Section 65B Indian Evidence Act** ke under 100% admissible hai [cite: 5].

Q6: On-the-spot Judge test scenario: Agar judge bole ki live 3-hop links filter karke dikhao, to kya karenge?

Ans: Humne backend me dynamic Neo4j Cypher Query integrate ki hai: `MATCH (p:Person {id:'X'})-[r:TRANSFERRED*1..3]->(target) WHERE r.amount > 500000 RETURN p, r, target` [cite: 5]. Yeh query 2 seconds me live graph update karke dikha deti hai [cite: 5].